Status_Summary

RK3588 / Orange Pi 5 Max — A/B-virt Emulator — Status Summary

Revision: 3 **Last modified:** 2026-06-11T13:15:00Z **Companion of:** `Status.md` (§11.4.56 two-audience parity).

Page 1 — For the operator (plain language)

We are building a way to test the RK3588 / Orange Pi 5 Max over-the-air (OTA) update flow **without needing the actual board on the desk yet** — by emulating the device on this Mac.

What works today:

- The emulated device **boots up and runs**. We built a small aarch64 Linux image and it boots all the way to a working login on this Mac (using Apple's built-in virtualization). We have the full boot recording as proof.
- **The heart of the A/B update now works and is proven.** Using a real bootloader (U-Boot 2024.01) on the Mac, the system can **switch from system half A to system half B** — we ran it and the device genuinely came up on the other half (confirmed from inside the running device), and we repeated it twice with identical results.
- **Automatic roll-back works too.** When we mark the new half as "bad" (too many failed boot attempts), the bootloader automatically **falls back to the known-good half** — we ran it and the device recovered to the good half. A second control run proved the roll-back only happens when a half is actually bad, not every time.
- The container-based test (control plane talking to a fake device) also works end-to-end.
- Our automated quality bank for this project passes its checks (12 of 12), including the check that the boot recording above is real and on file.

What is still pending (NOT done yet — honestly):

- **The dm-verity tamper-protection layer (RAUC) is NOT proven yet.** The script for it is written but has not been run — it needs a signed update bundle and a small reconciliation between the RAUC slot scheme and the bootloader slot scheme we just proved. Until we run it and capture the result, we do not claim it works.

- **The full real-Android A/B update test** needs a **Linux machine** (it cannot run on this Mac — the Mac lacks the required virtualization device). It is ready to run the moment your incoming Linux host is available; for now it is correctly **skipped**, not faked.
- Testing on the **real RK3588 / Orange Pi 5 Max board** is pending — no board on the bench yet.

Bottom line: boots on the Mac = proven; the real A/B slot switch and automatic roll-back = **proven** (with recordings on file). Still pending: dm-verity (RAUC), the Linux-host real-Android A/B test, and the real board. **No release version has been tagged** — that needs the full ladder green, including the Linux-host and real-board tiers, so tagging now would be overclaiming. We do not pretend anything works that has not been captured as evidence.

Page 2 — For software engineers

Fidelity ladder. T0 podman control-plane+emulator (SHIPPED, protocol-only) → T1 A/B-virt base image on QEMU `virt` + HVF (**GREEN**: live-userspace boot + real U-Boot 2024.01 A/B slot switch (E3) + corrupt-slot auto-rollback (E5) PROVEN; RAUC dm-verity (E4) AUTHORED, not yet run) → T2 Cuttlefish real Android `update_engine` A/B + AVB/dm-verity + auto-rollback incl. PWU-CF-2 corrupt-slot rollback (SKIP/UNVERIFIED on this host, Linux+KVM-gated) → T3 real RK3588 hardware (PENDING).

Proven (captured evidence):

- **PWU-AB-1 base image + real `u-boot.bin`** — `tests/emulator/ab_virt/out/images/Image` (MD5 `9f3670cd7dba7bdeeebe2c6d791e929a`) + `tests/emulator/ab_virt/out/images/rootfs.ext2` (MD5 `a056760e88eea575977be13e38cfe430`); builder `tests/emulator/ab_virt/build_image.sh`. The U-Boot+RAUC build COMPLETED — the E3/E5 verdict files record `U-Boot 2024.01 (Jun 11 2026 - 08:21:42 +0000)` driving the boot.
- **PWU-AB-1 foundation boot** — `docs/qa/20260611T061626Z-ab-virt-boot/console.log` (196 lines): kernel on Apple CPU MIDR `0x610f` (`physical CPU … [0x610f0000]`), Buildroot `buildroot login: root`, post-login `uname -a → Linux buildroot 6.1.44 … aarch64`, sentinel `HELIX_USERSPACE_LIVE_OK` (line 182), clean `poweroff`. §11.4.107 liveness battery (full transcript + post-login sentinel, not a single frame). Driver `tests/emulator/ab_virt/boot_smoke.sh`.
- **PWU-AB-1 A/B slot switch (E3)** — `docs/qa/20260611T094958Z-ab-slot-switch/`: `verdict.txt` (Run A rc=0 → `HELIX_SLOTID=A` `/dev/vda2`; Run B rc=0 → `HELIX_SLOTID=B` `/dev/vda3` — the slot SWITCHED), `consoleA.log` / `consoleB.log` transcripts, `determinism_n3.txt` (**3/3 identical PASS**, §11.4.50). Each run asserts the post-login `slot_id` sentinel + `findmnt` root-device + kernel `helix_slot=` + the negative “did NOT boot the other slot” check (real switch, not a no-op). Drivers `tests/emulator/ab_virt/ab_slot_switch.sh` + `uboot_ab/boot.cmd` (commit `18ed84a`).
- **PWU-AB-3 corrupt-slot auto-rollback (E5)** — `docs/qa/20260611T095918Z-ab-rollback/`: `verdict.txt` (Run ROLLBACK rc=0: head=bad slot B, `bootcount=2 > bootlimit=1` → altbootcmd swap → booted slot A `/dev/vda2`; Run CONTROL rc=0: head B, no rollback → B `/`

dev/vda3), `consoleROLLBACK.log` shows `A/B: bootcount=2 > bootlimit=1 -> rolling back (altbootcmd swap)` → `booting slot A` → guest `HELIX_SLOTID=A`. The CONTROL run is the negative proof the rollback fires only on a bad slot. Drivers `tests/emulator/ab_virt/ab_rollback.sh` + `uboot_ab/boot.cmd` (commit `42be557`).

- **T0 round-trip** — `tests/emulator/tier1_container_e2e.sh` (register → update-check → telemetry asserted from captured podman logs under `docs/qa/<run-id>/`).
- **HelixQA bank static gate (E9)** — `bash tools/helixqa/run_bank.sh --dry-run --bank tools/helixqa/banks/helix_ota.yaml` → 12 passed / 0 failed / 0 skipped. `HOTA-AB-VIRT-BOOT` declares evidence `docs/qa/20260611T061626Z-ab-virt-boot/console.log` (verified present). The runner FAILs any challenge whose declared evidence is missing — this gates that the boot evidence exists + is wired into the bank (it does NOT itself exercise A/B apply).

Pending / not proven (§11.4.6):

- **RAUC dm-verity slot integrity (E4)** = PENDING_FORENSICS. Driver `tests/emulator/ab_virt/ab_rauc_verity.sh` AUTHORED + parse-clean but **NOT run** — gated on a signed `.raucb` bundle + reconciling the RAUC slot-class scheme with the proven `boot.cmd` `BOOT_ORDER` env scheme. No dm-verity evidence yet.
- **T2 Cuttlefish (E7)** = SKIP / UNVERIFIED-pending-Linux-host. Driver `tests/emulator/tier2_cuttlefish_ab.sh` (PWU-CF-2 corrupt-slot auto-rollback section authored, mirrors ab_virt PWU-AB-3); host gate `/dev/kvm` absent on this Apple-Silicon macOS host (verified). Exit-3 honest topology SKIP per §11.4.3/§11.4.112. Script header marks the exact OTA-apply + rollback invocation `UNCONFIRMED:` (Virtual-A/B vs legacy A/B; `update_device.py` vs `cvd`) — runtime-detected, never guessed. Ready for the operator's incoming Linux + nested-KVM host.
- **T3 hardware (E8)** = PENDING_FORENSICS. No board.
- **No release tag** — §11.4.40 needs the full ladder GREEN (T2 + T3 still SKIP/PENDING); tagging now would be a bluff.

Provenance: `dd43738` (research + dev-host A/B-virt build infra) → `d5374d0` (PWU-AB-1 foundation GREEN, live-userspace boot) → `4278aa9` (U-Boot + RAUC + GPT tooling) → `0f120a6` (parallel streams: HelixQA bank wiring + these Status docs + Cuttlefish PWU-CF-2 rollback) → `6581ab4` (A/B disk assembler + U-Boot bootcount slot-select, AUTHORED) → `0d27438` (build host-tools `libssl-dev` fix) → `18ed84a` (**PWU-AB-1 REAL A/B slot switch PROVEN**, E3) → `afcad1d` (determinism soak + PWU-AB-2 `ab_rauc_verity.sh` / PWU-AB-3 authored) → `42be557` (**PWU-AB-3 REAL corrupt-slot AUTO-ROLLBACK PROVEN**, E5). HEAD `42be557`.

§-refs: §11.4.45 (Status doc), §11.4.56 (two-audience summary), §11.4.5 (captured-evidence table), §11.4.6 (no-guessing), §11.4.107 (liveness), §11.4.3 / §11.4.112 (topology-gated SKIP), §11.4.44 (revision header).